



What is Wrong with These Surveys?



Identify what is wrong in each of these surveys. Be sure to explain.

1. The mayor of Springfield is interested in finding out the average age of people in the city. He obtains a list of all the homeowners in the city and then contacts a simple random sample of 300 people. He uses the sample to estimate the average age of all the people in the city.

a. What is wrong with this survey?

Non-homeowners won't be included. ← undercoverage

b. Do you think the mayor will over or underestimate the true mean age of people in Springfield? Why?

Overestimate. Older people are more likely to be homeowners, while younger people are more likely to rent.

2. The principal at a school wants to know the proportion of students who did all their homework last night. She selects a simple random sample of 100 students and send an email to each of them asking if they did all their homework last night. Of the 40 responses, 36 of the students said that they did all their homework last night (90%).

a. What is wrong with this survey?

Only 40 students replied. 60 did not. ← nonresponse
Students might lie and say they did do homework. ← response bias

b. Do you think the administration will over or underestimate the true proportion of students who did all their homework last night? Why?

Overestimate. The students who did respond were proud they did their work or they lied and said they did it.

3. There is a proposal to cut some of the funding for the fire department in a city. One of the firefighters in the city wants to know the proportion of people in his neighborhood who support the budget cuts. He takes a random sample of 30 homes and visits them dressed in his uniform.

a. What is wrong with this survey?

The firefighter wore his uniform so the people knew he was a firefighter. ← response bias

b. Do you think the firefighter will over or underestimate the true proportion of his neighbors who support the budget cuts? Why?

Underestimate. People are more likely to lie and say they are against the budget cuts.

Lesson 2.4 – Potential Problems with Sampling

QuickNotes

LT#1 Undercoverage: some people are less likely to be chosen

Examples: calling landlines, surveying homeowners

Nonresponse: people can't be reached or refuse to answer

Examples: don't answer or hang up on phone call

LT#2 Response bias: problems in the data gathering instrument or process

Examples: people lie (self reported responses), wording of question

Check Your Understanding

1. A counselor at a large high school wants to learn about student opinion regarding remote learning versus in-person learning. Each of the following sampling methods introduces a possible source of bias. Name and explain the type of bias.
 - a. The counselor selects a random sample of students from those who signed up for remote learning.

Undercoverage, the students who are doing in-person learning are not included.

- b. The counselor selects a random sample of 100 students and sends them all an email, but only 28 students respond.

Nonresponse, there were 72 students who were chosen to be in the sample, but didn't respond. *not voluntary response

- c. The counselor surveys the first 50 students that submit their learning preference for the upcoming school year.

Convenience sample, there was no random sample in the process, so the sample may not represent the population.

2. Students at a certain high school completed a survey that included the following question:

Social media use can negatively affect teens, distracting them, disrupting their sleep, and exposing them to bullying, rumor spreading, unrealistic views of other people's lives and peer pressure. How many hours per week do you use social media?

What type of bias is present? Will the survey underestimate or overestimate the true average number of hours per week that students use social media? Explain.

Response bias due to wording of the question. The survey will likely underestimate the true average number of hours on social media because students won't be truthful.